

Antony Sikorski

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EDUCATION

Colorado School of Mines (CSM)

2022 - 2026

PhD, MS

- PhD in Statistics awarded July 2026
- Data Science MS awarded May 2024
- **Advisors:** [Douglas Nychka](#), [Daniel McKenzie](#)
- **Thesis:** Fast, Neural, and Sparse Methods for Big Spatial Data
- **Research Focus:** Combining machine learning and spatial statistics to create fast, accurate, and interpretable methods for big datasets.

University of California, San Diego (UCSD)

2018 - 2021

BS, Major in Applied Mathematics, Minor in Physics

AWARDS

NSF GRFP

Sept 2024

- Awarded the National Science Foundation Graduate Research Fellowship in the second year of my PhD.

Graduate Research Award

Apr 2026

- Awarded the 2026 Graduate Research Award for excellence and impact in research from the Applied Mathematics and Statistics Department at the Colorado School of Mines.

WORK EXPERIENCE

Meta

May 2025 - Present

Data Scientist

Aug 2026 - Present

- Developing AI models and driving automation to solve large-scale data problems.

Reality Labs – Data Scientist Intern

May 2025 - Aug 2025

- Formulated custom convex optimization procedures with cvxpy and pyomo in Python.
- Developed and implemented a suite of deep learning and time series forecasting models.
- Leveraged geospatial data to build custom tooling to aid leadership in key decisions.

NASA Jet Propulsion Laboratory

May 2023 - Aug 2023

Machine Learning Intern

- Developed an unsupervised anomaly detection system using adversarially trained auto-encoders to identify communication failure modes between spacecraft and the Deep Space Network (DSN) antenna fleet in multivariate time series data.
- Designed custom parsers and automated several previously manual data acquisition pipelines, leveraging LLMs to rapidly process the results. Routines that once occupied hours of engineers' time now occur in seconds.

Excelitas Technologies

Feb 2022 - Mar 2023

Data Consultant

Aug 2022 - Mar 2023

- Continued to support data engineering and analysis projects for the improvement of the Excelitas data layer.

Data Analytics Engineer

Feb 2022 - Aug 2022

- Identified critical to quality factors and worked with Process and Sustaining engineering teams to drive manufacturing improvements.
- Engineered measurement database to encapsulate all necessary BOM structure measurements.
- Automated the query of data to allow for statistical process control and process capability analysis.

Tutors and Friends

Jun 2020 - Feb 2022

Academic Tutor

- Tutored mathematics, physics, ACT, and SAT for all ages, with a focus on high school students and undergraduates.

RESEARCH EXPERIENCE

Predictive Models for Large Spatial Data (Thesis Projects)

Aug 2022 - Present

Colorado School of Mines

Advisors: [Douglas Nychka](#), [Daniel McKenzie](#)

- **Project 1:** Sped up the normalization of basis functions for fast Gaussian Process approximations. Two fast algorithms implemented in the [LatticeKrig R package](#).
- **Project 2:** Created the [LatticeVision](#) framework, in which image-to-image neural networks—both U-Net and transformer-based—are used for non-stationary parameter estimation on large spatial datasets.
- **Project 3:** Expanded upon and applied the LatticeVision framework to a transfer-learning setting, enabling amortized neural parameter estimation across spatial supports. Fused numerical model outputs with sparse monitoring station observations to model atmospheric pollution in Italy.

LEAP Momentum Fellowship

May 2024 - Aug 2024 (Present)

Columbia University, NYU Courant

Advisor: [Sara Shamekh](#)

- Supervised the research of two undergraduate students, resulting in their first conference presentation.
- Used deep learning and equation discovery methods to create interpretable parametrizations of vertical turbulent fluxes in the atmosphere. These advances improve the accuracy of long term climate model predictions.
- Led introductory workshops on data visualization, Jupyter notebooks, deep learning with PyTorch, and version control.

Classification of Political Entity Relationships

Jun 2021 - Feb 2022

University of California, San Diego

Advisor: [David Meyer](#)

- Focused on quantifying and classifying relationships between countries using data from the ICEWS dataset with the aim of applying structural balance theory to predict changes in power dynamic.

Synthesis of Strongly Correlated Quantum Materials

Jan 2019 - Nov 2019

University of California, San Diego

Advisor: [Brian Maple](#)

- Used tetra-arc and Optical Floating zone growths to synthesize superconductors and other strongly correlated quantum material in an attempt to find higher temperature superconductors.

Searching for Quantum Spin Liquids

Jun 2016 - Sept 2019

Colorado State University, Fort Collins

Advisor: [Kathryn Ross](#)

- Synthesized single crystals of strongly correlated materials in an attempt to study the exciting quantum spin liquid state. Used an Optical Floating Zone to grow crystals of Ytterbium Silicate and analyzed the structure through neutron scattering.

TEACHING EXPERIENCE

Introductory Statistics (Math 201)

Jan 2024 - May 2024

Colorado School of Mines

- Taught the introductory undergraduate statistics course covering hypothesis testing, confidence intervals, regression, and general data analysis and wrangling proficiency in R.

Differential Equations (Math 225)

Aug 2022 - May 2023

Colorado School of Mines

- Led recitations for two semesters and graded as a TA for the undergraduate differential equations course.

TECHNICAL SKILLS

Python: Generally used for machine learning and data analysis. Notable proficiencies outside classic libraries: PyTorch, TensorFlow, Keras, PySR, Scikit-learn, Pandas, Jupyter Notebooks

R: Used for data analysis and statistical modeling. Currently developing an update for significant computational efficiency in LatticeKrig package for large spatial data. Notable packages: fields, LatticeKrig, dplyr, ggplot2, RMarkdown

SQL: Used for querying large datasets: PostgreSQL, Presto, Spark

Typesetting: proficient with $\LaTeX$ and Markdown

Other: Proficiency in Git, CI/CD, Julia, Matlab, CI/CD.

Human Languages: English (fluent), Ukrainian (beginner), Russian (fluent), Spanish (beginner)

PUBLICATIONS

- “LatticeVision: Image to Image Networks for Modeling Non-Stationary Spatial Data”, **A. Sikorski**, M. Ivanitskiy, N. Lenssen, D. Nychka, D. McKenzie, *International Conference on Artificial Intelligence and Statistics (AISTATS)* (2026)
- “A Non-stationary, Amortized, Transfer Learning Approach for Modeling Italian Air Quality”, A. Fusta Moro*, **A. Sikorski***, D. McKenzie, A. Fassò, D. Nychka *Spatial Statistics* (2026) (* signifies equal contribution)
- “Normalizing Basis Functions: Approximate Stationary Models for Large Spatial Data” **A. Sikorski**, D. McKenzie, D. Nychka, *Stat* (2024)
- “Crystal Growth of Quantum Magnets in the Rare-Earth Pyrosilicate Family $R_2Si_2O_7$, (R= Yb, Er) Using the Optical Floating Zone Method” H. S. Nair, T. DeLazzer, T. Reeder, **A. Sikorski**, G. Hester, & K. A. Ross, *Crystals* (2019)

DISSERTATION

- “Fast, Neural, and Sparse Methods for Big Spatial Data”, **A. Sikorski**, PhD Dissertation, Colorado School of Mines (2026).
Advisors: D. Nychka, D. McKenzie.

SOFTWARE DEVELOPMENT

- LatticeKrig (R): Computationally efficient methods for fitting and predicting from large spatial datasets. Available on [CRAN](#).
- LatticeVision (Python): Methods for training, evaluating, and applying image-to-image neural estimators to big, non-stationary spatial data. Development version currently on [Github](#).

TALKS & PRESENTATIONS

- “LatticeVision: Image to Image Networks for Modeling Non-Stationary Spatial Data”. **A. Sikorski**, M. Ivanitskiy, N. Lenssen, D. Nychka, D. McKenzie, poster at the *29th International Conference on Artificial Intelligence and Statistics (AISTATS)* (2026)
- “Vision Models for Big, Non-Stationary, Spatial Data”. **A. Sikorski**, M. Ivanitskiy, D. McKenzie, D. Nychka
 - Talk at *GRADS (The Graduate Research and Discovery Symposium)* (2025)
 - Talk at *AMS GVD Research Symposium* (2025)
- “Machine Learning for Equation Discovery in Climate Science”, **A. Sikorski**, invited talk at *Mines Optimization and Deep Learning Seminar (MODL)* (2024)
- “Parametrizing Turbulent Fluxes in the Planetary Boundary Layer with Symbolic Regression”, L. Pong, G. VanZetten, **A. Sikorski**, Y. Qu, S. Shamekh, talk at *The New York Climate Exchange* (2024)
- “Basis For Change: Approximate Stationary Models for Large Spatial Data”, **A. Sikorski**, D. McKenzie, and D. Nychka
 - Talk and poster at *Extremes* (2024)
 - Poster at *JSM (Joint Statistical Meeting)* (2024)
 - Talk and poster at *IMSC (International Meeting on Statistical Climatology)* (2024)
 - Poster at *GRADS (The Graduate Research and Discovery Symposium)* (2024)
- “Fast Prediction and Parameter Estimation for Large Spatial Data with Machine Learning” **A. Sikorski**, invited talk at *AMS Graduate Student Colloquium* (2023)
- “Exploring Neural Likelihood Surfaces for Spatial Processes”, **A. Sikorski**, talk at *Mines Optimization and Deep Learning Seminar (MODL)* (2023)
- “Optical Floating Zone Growth of $Yb_2Si_2O_7$ ”, **A. Sikorski**, H.S. Nair, T. Reeder and K.A. Ross, presented at the *APS Four Corners Conference* (October 2017)

SERVICE

Undergraduate Students Mentored

- Laura Pong (LEAP, 2024)
- Greta VanZetten (LEAP, 2024)

Organization

- Session Chair: “Risk Analysis, Management, and Investing” (JSM, 2024)
- Department Representative: Mines Graduate Student Government (GSG)

Software Review

- darkroast R package (2024)
- ZANJ Python package (2023)

Affiliations

- NASA Jet Propulsion Laboratory (JPL)
- Mines Optimization and Deep Learning (MODL)
- Learning the Earth with Artificial Intelligence and Physics (LEAP) NSF Science and Technology Center (STC)
- Society for Women in Mathematics (SWiM)
- American Statistical Association (ASA)